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MCR DIMM Promotion Meeting

Dec 2022

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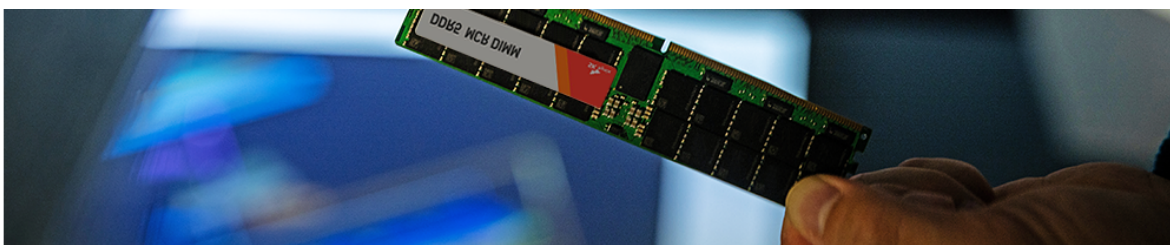
Interview With the SK hynix Team Behind MCR DIMM, "the Best Server DRAM"

#DDR5 #MCRDIMM #server DRAM

January 17, 2023 | SK hynix

AI Summary ▾





SK hynix once again proved to be the first and the best when it comes to innovations as it became the world's first company to [successfully develop working samples of the DDR5 MCR DIMM](#)^{*}, a server DRAM product, in December 2022. This successful development of a sample significantly enhances the performance of the server DRAM. The biggest improvement is that the MCR DIMM breaks away from the traditional idea of focusing solely on the operating speed of individual DRAM units to boost the speed of DDR5. Instead, it adds a customized module to individual DRAM units to increase the speed. The operating speed of the MCR DIMM is over 8 Gbps (gigabits per second), which is at least 80% faster than existing server DRAMs that have a speed of 4.8 Gbps. This makes it the fastest server DRAM in the world.

^{*}MCR DIMM (Multiplexer Combined Ranks Dual In-line Memory Module): A module product with multiple DRAMs bonded to a motherboard, in which two ranks^{*} (basic information processing units) operate simultaneously, resulting in improved speed.

^{*}Rank: A set of basic data transfer units exported from the DRAM module to the CPU. Usually, 64 bytes of data are transferred to the CPU as a unit.

As SK hynix sees the server memory market as pivotal to overcoming the recent downturn of the semiconductor industry, the company has focused on developing technologies related to server DRAMs. The SK hynix team who achieved innovation through new methods and paths emphasized the importance of challenging oneself, saying: "No matter how difficult it is, opening up new paths is always worthwhile." This was the motivation for the MCR DIMM team members, Kim Hong Bae, Kim Yeong Jun, and Yi Jong Yun, who committed themselves to developing the leading server DRAM.





▲ Figure 1. The members behind the development of the MCR DIMM—Kim Hong Bae (left), Yi Jong Yun (middle), and Kim Yeong Jun (right)—discussing the server DRAM's development process.

Server DRAMs are Essential for Future Digital Era's Infrastructure

Today, society is undergoing drastic changes. Almost every piece of information is generated, recorded, and shared digitally. This is what's called digital transformation (DT), and the server DRAM is a crucial component of the infrastructure that builds this digital world. Therefore, SK hynix's development of the MCR DIMM is meaningful in many ways.

The project head, Professional Leader Kim Hong Bae at DRAM Product Planning, emphasized the role of the server DRAM. "You need wider highways and faster cargo vehicles if you want to efficiently transport shipments. We call this logistics infrastructure," Kim said. "The role of server DRAMs is similar to that of a cargo vehicle. It helps to move data faster and smoother. That is why I think a server DRAM is the core component of the digital world's infrastructure."

He continued: "Looking at it from an infrastructure point of view, the importance of the MCR DIMM is considerable. A DRAM that can process data more than 80% faster than existing DRAMs and in a much more stable way has been developed to address the exponentially increasing amount of data nowadays. And considering these performance improvements, this is truly an extraordinary achievement."





“The role of server DRAMs is similar to that of a cargo vehicle. It helps move data faster and smoother.”

– Professional Leader Kim Hong Bae, DRAM Product Planning

▲Figure 2. “The role of server DRAMs is similar to that of a cargo vehicle. It helps move data faster and smoother,” said Professional Leader Kim Hong Bae, DRAM Product Planning.

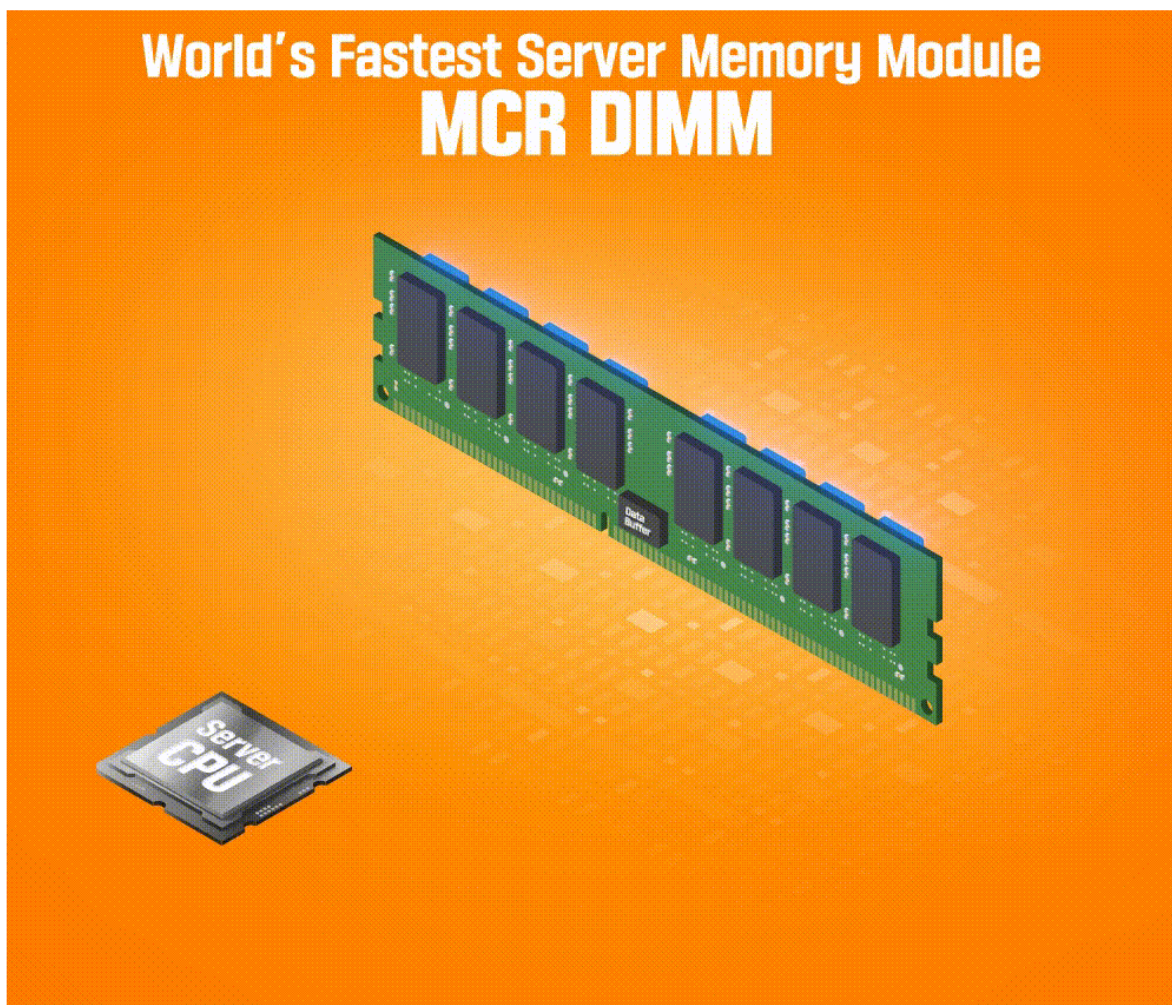
Kim Yeong Jun, a technical leader at DRAM Product Planning, placed great importance on being able to respond to the rapid growth of data that will occur in the near future. “I think the amount of data we will create down the road will increase tremendously. Therefore, the amount of data that DRAMs need to process will also surely increase. What would happen if every household has an 8K TV? In order to supply 8K-level streaming to every home, media servers will have to process an enormous amount of data. It is inevitable that we will need hardware that can handle more data,” Kim said. “The problem is that it will not only be the video field that will develop in such a way. We are currently facing an era of digital transformation where all areas including finance, shopping, travel, culture, and transportation are being digitalized. Eventually, every aspect of an individual’s life will generate data, so servers that process this data will require higher-level systems and hardware. I can confidently say that SK hynix has prepared itself for these future demands.”

Technical Leader Yi Jong Yun at DRAM AE (Application Engineering), who conducted the MCR DIMM test, also showed pride in the successful development. “SK hynix has continued to show a strong presence in the server DRAM market, and the success of the MCR DIMM’s development served as an opportunity to declare once again that our company produces the best server DRAMs,” Yi said. “As my colleagues already mentioned, I am looking forward to the future that the MCR DIMM will make.”

SK hynix Pushes Technological Limits to Lead the Server DRAM Market

The MCR DIMM represents a leap forward in performance, typically unseen in the DRAM sector. It is common to see an operating speed increase of about 800 Mbps (Megabits per second) when developing a new generation of the DDR5 DRAM. For example, if the DRAM offered an operating speed of 4,800 Mbps, the next generation is expected to have a speed of approximately 5,600 Mbps. However, SK hynix made yet another breakthrough by developing the MCR DIMM that boasts an operating speed of over 8,000 Mbps.

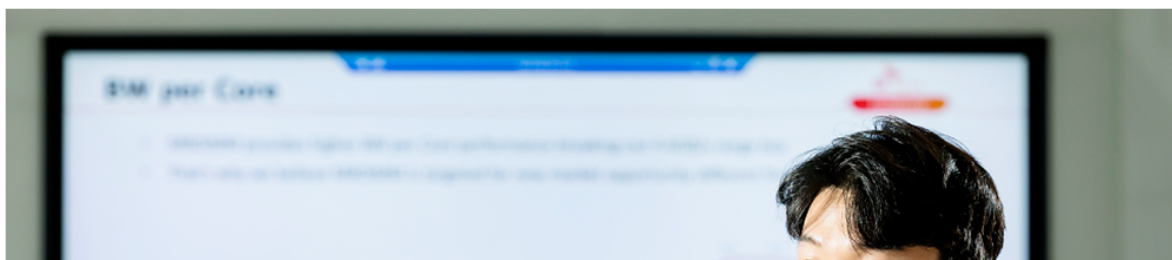
World’s Fastest Server Memory Module
MCR DIMM

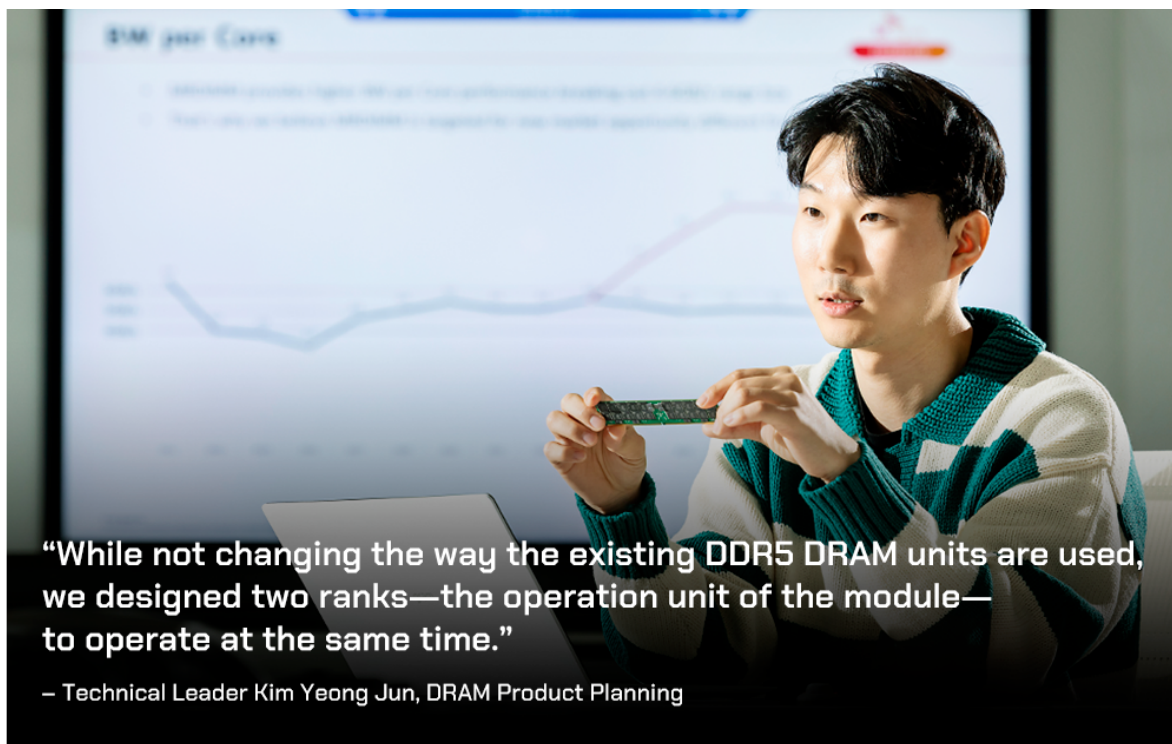


▲ Figure 3. The operation structure of the world's fastest server DRAM, the MCR DIMM.

Kim Hong Bae had more to add on the performance innovations of the MCR DIMM. "In comparison to server DRAMs that had a fixed rate of performance enhancement for each generation, server CPUs (Central Processing Unit) were expected to have a performance enhancement rate that was higher than that of DRAMs," Kim said. "It was critical to prevent a considerable difference between the CPU's operating speed and the DRAM's data processing speed, as the DRAM could eventually cause a bottleneck*. So, we achieved a rapid improvement in the MCR DIMM's performance by making new and unprecedented innovations."

*Bottleneck: Originally a term to describe traffic disruption typically caused by vehicles waiting for signals or entering narrower roads. In the computer industry, a bottleneck refers to the situation when one component reaches its maximum capability and restricts the performance of other components in the system.





“While not changing the way the existing DDR5 DRAM units are used, we designed two ranks—the operation unit of the module—to operate at the same time.”

– Technical Leader Kim Yeong Jun, DRAM Product Planning

▲Figure 4. “While not changing the way the existing DDR5 DRAM units are used, we designed two ranks—the operation unit of the module—to operate at the same time,” said Technical Leader Kim Yeong Jun at DRAM Product Planning, regarding the MCR DIMM.

Kim Yeong Jun provides a behind-the-scenes explanation of the newly attempted techniques that went into the product development. “In the process of developing the MCR DIMM, we introduced a new concept to increase operating speeds,” he explained. “While not changing the way the existing DDR5 DRAM units are used, we designed two ranks—the operation unit of the module—to operate at the same time. While 64 bytes of data are transmitted in one rank, 128 bytes of data can be transmitted at once by operating two ranks simultaneously. A data buffer* is used for this operation as it combines the 64 bytes of data transmitted in two ranks and creates the collective transmit of 128 bytes.”

He added: “In particular, collaboration with global companies—Intel (U.S.) and Renesas (Japan)—played an important role in this development.” Intel, which develops server CPUs, and Renesas, which develops data buffers for the MCR DIMM, collaborated with SK hynix to create synergy.”

*Buffer: A component that is mounted on top of the DRAM module to optimize the performance of the signal transmission between the DRAM and the CPU. It is mainly installed in server DRAM modules that require high performance and stability.

A Challenge that Others Avoided Became a Chance to Innovate

The development of the MCR DIMM was especially difficult because everything from the design to the testing process consisted of methods that no one had ever tried before.

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The development of the MCR DIMM was especially difficult because everything from the design to the testing process consisted of methods that no one had ever tried before. Since it was a distinct product developed without a standard, every part of the process was a challenge.



**“In the case of the MCR DIMM,
it was not easy to build a completely new test environment.”**

– Technical Leader Yi Jong Yun, DRAM AE

▲ Figure 5. “In the case of the MCR DIMM, it was not easy to build a completely new test environment,” said Technical Leader Yi Jong Yun, DRAM AE, recalling the development of the groundbreaking test program for the MCR DIMM while holding a sample.

The team members who developed the product described the biggest challenges they faced during the whole process. “In the case of the MCR DIMM, it was not easy to build a completely new test environment,” Yi Jong Yun said. “In fact, when developing a new generation of current products, there is a target for performance level. In the case of DDR, the industry expects a performance increase of 800 Mbps for new generations. However, it was difficult to set a target since there were no references or related information in the MCR DIMM’s case. Developing cannot be done with only the concept; it needs a specific target as well. Accordingly, it was difficult to create a test program and environment to bring our concept into life. I am proud that SK hynix was able to overcome these hurdles and set up the foundation to lead in the server DRAM market by successfully developing the MCR DIMM.”

As for Kim Yeong Jun, he recalls initially proposing the product to the company. “The MCR DIMM is a new concept that has never existed before, so I remember working hard to convince colleagues on why we should develop this product and why clients would want to buy it. We needed to convince others that it was worth breaking away from the existing DRAM development methods to go with a new development method which is a risk.” Kim

As for Kim Yeong Jun, he recalls initially proposing the product to the company. “The MCR DIMM is a new concept that has never existed before, so I remember working hard to convince colleagues on why we should develop this product and why clients would want to buy it. We needed to convince others that it was worth breaking away from the existing DRAM development methods to go with a new development method which is a risk.” Kim said. “Ultimately, we are still defining the MCR DIMM market since it is not fully solidified yet. Since the project has not been completely finalized and as there is still time left until its mass production, greater efforts are needed in the future. When the development and mass production of MCR DIMMs are completed and widely adopted in the market, I am really looking forward to the customers’ reactions and the changes that will occur in the server DRAM industry.”

Kim Hong Bae also expressed his anticipation for the wide use of MCR DIMMs down the road. “We are realizing how important data centers and servers are these days. And through our experiences, we know that if servers do not function, many aspects in our daily life stop. In fact, server facilities are a key piece of modern-day infrastructure,” Kim said. “For those of us who have become accustomed to the digital world, data centers and servers have become indispensable, and SK hynix is committed to building better data centers and servers for everyone. I hope we can contribute to creating a better future by developing more innovative products, starting with the MCR DIMM.”

While the members take much pride in the fact that their product has improved performance by more than 80% compared to existing products, they admit that there is still much work to be done. As they continue to develop the MCR DIMM, the team members of SK hynix, who have achieved innovation by tackling challenges that only a few have ever dared to try, will not shy away from taking the road less traveled in the future as well.

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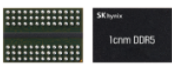
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